



Multithreading & processing

Multi-threading and Multi-Processing

1. Program

↳ Is a sequence of instructions written in programming language [eg: Python, C++]

Eg: Google Chrome → exc ← Program → Browser should work

2. Process

↳ It is an instance of a program that is being executed.

Eg: Opening Google Chrome Program (Task Manager)

Excel - process - id

Email - process - id

* I/O Requirements

↓
Increased exec time to switch between process

Code Segment	Data Segment	Heap
Stack	Register	
S-thread		

Seperate memory space

* One process cannot corrupt another process

3. Threads

↳ Is a unit of execution within a process.

Single Threaded Process

Eg: MS PAINT → 



Multi-Threaded Process

Multiple threads running concurrently